Build a timeline you can edit

A practical Codex workflow for native Fusion modeling, with specifications, guarded steps and measured edit checks.

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The target is simple: fix a dimension or feature without rebuilding the whole part. A useful timeline must preserve native sketches, parameter expressions and the intended dependencies. A healthy-looking solid is only part of the result.

What this integration combines

Input	Contribution
uwu's supplied prototype	Specification-first planning; defined interfaces and connections; one verified step at a time; editability as an acceptance goal.
Existing Fusion Guard	Codex gate and restricted scripts; before/after inspection; per-call local report verification; manual recovery checks.
Web and API research	Native timeline semantics, available MCP tools, and AutoTimeline as an optional imported-solid recovery path.

The working sequence

Specify > inspect > parameters > constrained sketch > native feature > accept > edit and measure.

Use the part-spec template to define how each part attaches, which dimensions may change, and what measurements will demonstrate success. This addresses missing anchors, threads, holder connections and fit requirements that hooks cannot infer.

The Claude prototype is reference material. Its hook registration, stdout reports, reset command and automatic flex test are not stacked onto the Codex implementation. The active project was left unchanged while this distribution was assembled.

Install, then prove the hook path

Use a separate project and a disposable parametric Fusion design for initial qualification.

1. Check compatibility. Windows, PowerShell and Python 3.11+ are required. Codex and Fusion must share access to local project files. The execute tool must be **mcp__fusion__fusion_mcp_execute**, accepting **featureType: "script"**, **object.script** and **object.readOnly**.

2. Run local checks. From the extracted kit root:

```
python -B fusion_guard/test_guard.py
python -B -m unittest discover -s tests -v
```

3. Copy to a new pilot project. Copy the runtime folder, specification template and relevant docs. Do not overwrite an existing guard, its policy or state. Keep the Python installation and project location stable.

4. Prepare and inspect the generated hook commands. Run the first command, review **fusion_guard/hooks.prepared.json**, then run the second:

```
python -B fusion_guard/install.py
python -B fusion_guard/install.py --install
```

5. Review trust in Codex. Review the project layer and PreToolUse, PostToolUse and Stop hooks. The installer preserves unrelated hook entries and backs up replaced local files. It does not grant trust or activate policy.

6. Deliberately enable the pilot. Stop all other Fusion writers first. The release policy defaults to disabled.

```
python -B fusion_guard/manage.py enable --reason "Pilot qualification"
```

7. Run the unchanged probe through Fusion. Send the full contents of **fusion_guard/probe.py** as the script with **readOnly: true**. Require accepted feedback and the matching local inspection report. The probe deliberately fails when the hook has not injected inspection.

Installed files, enabled policy and trusted settings are distinct from a verified live path. If the probe fails, diagnose loading and repeat it; do not edit the probe or mutate a model to test activation.

Measure the edit, not just the health

Build one small native plate, then change its length while preserving the existing feature sequence.

Parameter	Starting expression	Intent
plate_length	80 mm	Overall X length
plate_width	50 mm	Overall Y width
plate_thickness	4 mm	Extrusion distance
edge_offset	10 mm	Hole center to each outer edge
hole_diameter	5 mm	Four through-holes

Create a centered, fully constrained rectangle; one native extrusion; a fully constrained four-circle sketch; and one through cut. Create parameters, each sketch and each feature in separate accepted calls. Use named expressions for dimensions.

Change plate_length from 80 mm to 96 mm

Check	Required outcome
Feature history	Zero timeline growth; same intended native feature sequence; no duplicate replacement geometry.
Outer size	96 x 50 x 4 mm; still one solid.
Hole centers	X = +/-38 mm and Y = +/-15 mm. The 10 mm edge offset follows the outer edges.
Hole size and health	Diameter remains 5 mm. Changed sketches stay fully constrained. No new warnings or errors.

Record actual measurements with a tolerance, such as 0.01 mm for this simple pilot. Convert Fusion API internal length units from centimeters when reporting millimeters. The full procedure is in **docs/ACCEPTANCE.md**.

After a successful trial, restore 80 mm through a separate deliberate edit and remeasure. This is a planned test in a disposable design. If any step fails, stop modifications and follow manual recovery; do not run an automatic reset or rollback.

For a real part: replace these plate assertions with the required attachment, thread, anchor, fit and interference checks. A feature can be healthy while the mechanical design is wrong.

Share a reviewable starter kit

Send the versioned ZIP and this quick-start PDF. Use a repository and versioned releases for future updates.

Included	Use
Runtime + installer	Generate recipient-local hook paths; inspect source before trusting hooks.
Specification + pilot	Define interfaces and expected edit behavior before modeling.
Tests + validation record	Separate local software checks from live Fusion qualification.
Research guide	22 primary sources and comparison material; native timeline modeling remains the main route.

Evidence and remaining work

Completed: 38 retained runtime tests and 12 installer regression tests pass using local fixtures and disposable filesystem copies. The prototype's 27 tests also passed during review. No Fusion model was modified for packaging.

Still required for each installation: real read-only probe, live native-model pilot, measured parameter edit and human review of mechanical interfaces. This release does not claim those recipient-specific checks have passed.

Boundaries to keep visible

The guard checks a restricted script path, not every Fusion action. GUI operations, other clients, alternate tool names and cloud sessions are outside its full inspection path. Non-script tool routes have different handling. Static checks are not a Python security sandbox.

Timeline marker movement, rollTo setters, reordering and grouping are blocked by this version. Automatic parameter perturbation and Undo/Redo are not enabled. Acceptance reports are state summaries, not full document backups or proofs of design intent.

Keep generated reports, CAD files, document identifiers, installation backups and machine-specific hook commands out of shared releases. The kit includes a clean source manifest and a ready-to-copy announcement.

Primary references

Codex hooks | Fusion TimelineObject | Fusion computeAll | Generate Timeline